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ELECTROMAGNETIC INTERFERENCE FILTER FOR ELECTRIC VEHICLE TRACTION INVERTER

Abstract

An inverter system controller for an automotive vehicle includes a housing and has disposed within the housing a Y-capacitor defining a terminal, a ground busbar, and a surface mount resistor in direct contact with the terminal and ground busbar to form an electromagnetic filter that lacks a printed circuit board.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an inverter for an electric vehicle (EV). More specifically the present disclosure relates to an electromagnetic interference (EMI) filter for an EV inverter.

BACKGROUND

[0002] Electric vehicles rely on one or more power inverters to convert electric energy between direct-current (DC) and alternating-current (AC) forms for supplying power to an electric machine for propulsion. During the operation, the inverter may emit EMI.

SUMMARY

[0003] A vehicle comprises an electric machine for propelling the vehicle, a battery for supplying electric power to the electric machine, and an electromagnetic interference filter. The electromagnetic interference filter is connected between the electric machine and battery, and includes a housing, a capacitor disposed within and insulated from the housing, a ground busbar attached to the housing, a resistor having a first terminal connected to a terminal of the capacitor and a second terminal connected to the ground busbar, and an insulating cover at least partially covering a top surface of the capacitor.

[0004] An automotive system comprises automotive power electronics including an electromagnetic interference filter, that lacks a printed circuit board, disposed within a housing of an inverter system controller.

[0005] An automotive component comprises an inverter system controller including a housing and having disposed within the housing a Y-capacitor defining a terminal, a ground busbar, and a surface mount resistor in direct contact with the terminal and ground busbar to form an electromagnetic filter that lacks a printed circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a diagram of an electrified vehicle illustrating drivetrain and energy storage components including an electric machine.

[0007] FIG. **2** is a circuit diagram of a power inverter and an EMI filter for an electric machine.

[0008] FIG. **3** is a design diagram of an EMI filter.

[0009] FIG. **4**A is a design diagram showing an enlarged portion of an insulating cover.

[0010] FIG. **4**B is another design diagram showing an enlarged portion of an insulating cover.

[0011] FIG. **5** is a design diagram of the EMI filter showing an enlarged portion of another embodiment.

DETAILED DESCRIPTION

[0012] Embodiments are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale. Some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art.

[0013] Various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0014] The present disclosure, among other things, proposes an EMI filter design for an EV power inverter.

[0015] FIG. **1** depicts an electrified vehicle **112** that may be referred to as a plug-in hybrid-electric vehicle (PHEV). A plug-in hybrid-electric vehicle **112** may comprise one or more electric machines **114** mechanically coupled to a hybrid transmission **116**. The electric machines **114** may be capable of operating as a motor or a generator. In addition, the hybrid transmission **116** is mechanically coupled to an engine **118**. The hybrid transmission **116** is also mechanically coupled to a drive shaft **120** that is mechanically coupled to the wheels **122**. The electric machines **114** can provide propulsion and braking capability when the engine **118** is turned on or off. The electric machines **114** may also function as generators and can provide fuel economy benefits by recovering energy that would normally be lost as heat in a friction braking system. The electric machines **114** may also reduce vehicle emissions by allowing the engine **118** to operate at more efficient speeds and allowing the hybrid-electric vehicle **112** to be operated in electric mode with the engine **118** off under certain conditions. An electrified vehicle **112** may also be a battery electric vehicle (BEV). In a BEV configuration, the engine **118** may not be present.

[0016] A traction battery or battery pack **124** stores energy that can be used by the electric machines **114**. The vehicle battery pack **124** may provide a high voltage direct current (DC) output. The traction battery **124** may be electrically coupled to one or more power electronics modules **126** (also referred to as a traction inverter). One or more contactors **142** may isolate the traction battery **124** from other components when opened and connect the traction battery **124** to other components when closed. The power electronics module **126** is also electrically coupled to the electric machines **114** and provides the ability to bi-directionally transfer energy between the traction battery **124** and the electric machines **114**. For example, a traction battery **124** may provide a DC voltage while the electric machines **114** may operate with a three-phase alternating current (AC) to function. The power electronics module **126** may convert the DC voltage to a three-phase AC current to operate the electric machines **114**. In a regenerative mode, the power electronics module **126** may convert the three-phase AC current from the electric machines **114** acting as generators to the DC voltage compatible with the traction battery **124**.

[0017] The vehicle **112** may include a variable-voltage converter (VVC) (not shown) electrically coupled between the traction battery **124** and the power electronics module **126**. The VVC may be a DC/DC boost converter configured to increase or boost the voltage provided by the traction battery **124**. By increasing the voltage, current requirements may be decreased leading to a reduction in wiring size for the power electronics module **126** and the electric machines **114**. Further, the electric machines **114** may be operated with better efficiency and lower losses.

[0018] In addition to providing energy for propulsion, the traction battery **124** may provide energy for other vehicle electrical systems. The vehicle **112** may include a DC/DC converter module **128** that converts the high voltage DC output of the traction battery **124** to a low voltage DC supply that is compatible with low-voltage vehicle loads. An output of the DC/DC converter module **128** may be electrically coupled to an auxiliary battery **130** (e.g., 12V battery) for charging the auxiliary battery **130**. The low-voltage systems may be electrically coupled to the auxiliary battery **130**. One or more electrical loads **146** may be coupled to the high-voltage busbar/rail. The electrical loads **146** may have an associated controller that operates and controls the electrical loads **146** when appropriate. Examples of the electrical loads **146** may be a fan, an electric heating element and/or an air-conditioning compressor.

[0019] The electrified vehicle **112** may be configured to recharge the traction battery **124** from an external power source **136**. The external power source **136** may be a connection to an electrical outlet. The external power source **136** may be electrically coupled to a charger or electric vehicle supply equipment (EVSE) **138**. The external power source **136** may be an electrical power distribution network or grid as provided by an electric utility company. The EVSE **138** may provide circuitry and controls to manage the transfer of energy between the power source **136** and the vehicle **112**. The external power source **136** may provide DC or AC electric power to the EVSE **138**. The EVSE **138** may have a charge connector **140** for plugging into a charge port **134** of the vehicle **112**. The charge port **134** may be any type of port configured to transfer power from the EVSE **138** to the vehicle **112**. The charge port **134** may be electrically coupled to a charger or on-board power conversion module **132**. The power conversion module **132** may condition the power supplied from the EVSE **138** to provide the proper voltage and current levels to the traction battery

124. The power conversion module **132** may interface with the EVSE **138** to coordinate the delivery of power to the vehicle **112**. The EVSE connector **140** may have pins that mate with corresponding recesses of the charge port **134**. Alternatively, various components described as being electrically coupled or connected may transfer power using a wireless inductive coupling.

[0020] Electronic modules in the vehicle **112** may communicate via one or more vehicle networks. The vehicle network may include a plurality of channels for communication. One channel of the vehicle network may be a serial bus such as a Controller Area Network (CAN). One of the channels of the vehicle network may include an Ethernet network defined by the Institute of Electrical and Electronics Engineers (IEEE) 802 family of standards. Additional channels of the vehicle network may include discrete connections between modules and may include power signals from the auxiliary battery **130**. Different signals may be transferred over different channels of the vehicle network. For example, video signals may be transferred over a high-speed channel (e.g., Ethernet) while control signals may be transferred over CAN or discrete signals. The vehicle network may include any hardware and software components that aid in transferring signals and data between modules. The vehicle network is not shown in FIG. **1** but it may be implied that the vehicle network may connect to any electronic module that is present in the vehicle **112**. A vehicle system controller (VSC) **148** may be present to coordinate the operation of the various components.

[0021] The electric machines **114** may be coupled to the power electronics module **126** via one or more conductors that are associated with each of the phase windings. Referring to FIG. **2**, a diagram of a circuit **200** including a power inverter and an EMI filter for an electric machine of one embodiment of the present disclosure is illustrated. With continuing reference to FIG. **1**, the vehicle **112** may include one or more power electronics controllers **201** configured to monitor and control the components of the power electronics module **126**. The power electronics controllers **201** may be under a global control or coordination of the VSC **148**. Further coordinated by the VSC **148** may be the main contactor **142** connected between the power electronics module **126** and the traction battery **124**. As illustrated in the present example, the main contactor **142** may be connected on a positive high-voltage (HV) busbar (HV rail, DC busbar or DC rail) **152***a*. Under normal discharge and regenerative operating conditions, the main contactor **124** may be closed by the VSC **148** to connect the traction battery **124** to the rest of the circuit allowing the traction battery **124** to be discharged or charged.

[0022] The conductors may be part of a wiring harness between the electric machine **114** and the power electronics module **126**. A three-phase electric machine **114** may have three conductors coupled to the power electronics module **126**. The power electronics module **126** may be configured to switch the positive HV busbar **152***a* and a negative HV busbar **152***b* to phase terminals of the electric machines **114**. The power electronics module **126** may be controlled to provide pulse-width modulated (PWM) voltage and sinusoidal current signals to the electric machine **114**. The frequency and/or duty ratio of the signals may be proportional to the rotational speed of the electric machine **114**. The controller **201** may be configured to adjust the voltage and current output of the power electronics module **126** at a predetermined switching frequency. The switching frequency may be the rate at which the states of switching devices within the power electronics module **126** are changed.

[0023] The power electronics module **126** may interface with a position/speed feedback device **202** that is coupled to the rotor of the electric machine **114**. For example, the position/speed feedback device **202** may be a resolver or an encoder. The position/speed feedback device **202** may provide signals indicative of a position and/or speed of the rotor of the electric machine **114**. The power electronics **126** may include a power electronics controller **201** that interfaces to the speed feedback device **202** and processes signals from the speed feedback device **202**. The power electronics controller **201** may be programmed to utilize the speed and position feedback to control the power electronics module **126** to operate the electric machine **114**.

[0024] The traction inverter or power electronics module **126** may include power switching circuitry **240** that includes a plurality of switching devices **210**, **212**, **214**, **216**, **218**, **220**. The switching devices **210**, **212**, **214**, **216**, **218**, **220** may be Insulated Gate Bipolar Transistors (IGBT), Metal Oxide Semiconductor Field Effect Transistors (MOSFET), or other solid-state switching devices. The switching devices **210**, **212**, **214**, **216**, **218**, **220** may be configured to selectively couple the positive HV busbar **152***a* and the negative HV busbar **152***b* to each phase terminal or leg (e.g., labeled U, V, W) of the electric machine **114**. The power electronics module **126** may be configured to provide a U-phase voltage, a V-phase voltage, and a W-phase voltage to the electric machine **114**. Each of the switching devices **210**, **212**, **214**, **216**, **218**, **220** within the power switching circuitry **240** may have an associated diode **222**, **224**, **226**, **228**, **230**, **232** connected in parallel to provide a path for inductive current when the switching device is in a non-conducting state. Each of the switching devices **210**, **212**, **214**, **216**, **218**, **220** may have a control terminal for controlling operation of the associated switching device. The control terminals may be electrically coupled to the power electronics controller **201**. The power electronics controller **201** may include associated circuitry to drive and monitor the control terminals. For example, the control terminals may be coupled to the gate input of the solid-state switching devices.

[0025] A phase leg of the inverter **126** may include switching devices and circuitry configured to selectively connect a phase terminal of the electric machine **114** to each terminal of the HV busbar **152**. A first switching device **210** may selectively couple the positive HV busbar **152***a* to a first phase terminal (e.g., U) of the electric machine **114**. A first diode **222** may be coupled in parallel to the first switching device **210**. A second switching device **212** may selectively couple the negative HV busbar **152***b* to the first phase terminal (e.g., U) of the electric machine **114**. A second diode **224** may be coupled in parallel to the second switching device **212**. A first inverter phase leg may include the first switching device **210**, the first diode **222**, the second switching device **212**, and the second diode **224**.

[0026] A third switching device **214** may selectively couple the positive HV busbar **152***a* to a second phase terminal (e.g., V) of the electric machine **114**. A third diode **226** may be coupled in parallel to the third switching device **214**. A fourth switching device **216** may selectively couple the negative HV-busbar **152***b* to the second phase terminal (e.g., V) of the electric machine **114**. A fourth diode **228** may be coupled in parallel to the fourth switching device **216**. A second inverter phase leg may include the third switching device **214**, the third diode **226**, the fourth switching device **216**, and the fourth diode **228**.

[0027] A fifth switching device **218** may selectively couple the positive HV busbar **152***a* to a third phase terminal (e.g., W) of the electric machine **114**. A fifth diode **230** may be coupled in parallel to the fifth switching device **218**. A sixth switching device **220** may selectively couple the negative HV busbar **152***b* to the third phase terminal (e.g., W) of the electric machine **114**. A sixth diode **232** may be coupled in parallel to the sixth switching device **220**. A third inverter phase leg may include the fifth switching device **218**, the fifth diode **230**, the sixth switching device **220**, and the sixth diode **232**.

[0028] The power switching devices **210**, **212**, **214**, **216**, **218**, **220** may include two terminals that handle the high-power current through the power switching device. For example, an IGBT includes a collector (C) terminal and an emitter (E) terminal and a MOSFET includes a drain terminal (D) and a source(S) terminal. The power switching devices **210**, **212**, **214**, **216**, **218**, **220** may further include one or more control inputs. For example, the power switching devices **210**, **212**, **214**, **216**, **218**, **220** may include a gate terminal (G) and a Kelvin source/emitter (K) terminal. The G and K terminals may comprise a gate loop to control the power switching device.

[0029] The power electronics module **126** may be configured to flow a rated current and have an associated power rating. Some applications may demand higher power and/or current ratings for proper operation of the electric machine **114**. The power switching circuitry **240** may be designed to include power switching devices **210**, **212**, **214**, **216**, **218**, **220** that can handle the desired power/current rating. The desired power/current rating may also be achieved by using power switching devices that are electrically coupled in parallel. Power switching devices may be electrically coupled in parallel and controlled with a common control signal so that each power switching device flows a portion of the total current flowing to/from the load.

[0030] The power electronics controller **201** may be programmed to operate the switching devices **210**, **212**, **214**, **216**, **218**, **220** to control the voltage and current applied to the phase windings of the electric machine **114**. The power electronics controller **201** may operate the switching devices **210**, **212**, **214**, **216**, **218**, **220** so that each phase terminal is coupled to only one of the positive HV busbar **152***a* or the negative HV busbar **152***b* at a particular time.

[0031] Various motor control algorithms and strategies are available to be implemented in the power electronics controller **201**. The power electronics module **126** may also include current sensors **204**. The current sensors **204** may be inductive or Hall-effect devices configured to generate a signal indicative of the current passing through the associated circuit. In some configurations, two current sensors **204** may be utilized and the third phase current may be calculated from the two measured currents. The controller **201** may sample the current sensors **204** at a predetermined sampling rate. Measured values of the phase currents of the electric machine **114** may be stored in controller memory for later computations.

[0032] The power electronics module **126** may include one or more voltage sensors. The voltage sensors may be configured to measure an input voltage to the power electronics module **126** and/or one or more of the output voltages of the power electronics module **126**. The power electronics module **126** may include a line voltage sensor **250** that is configured to measure a line voltage across the V and W phase outputs. The voltage may be a voltage difference between the V-phase voltage and the W-phase voltage. The voltage sensors may be resistive networks and include isolation elements to separate high-voltage levels from the low-voltage system. In addition, the power electronics module **126** may include associated circuitry for scaling and filtering the signals from the current sensors **204** and the voltage sensors. In some configurations, each phase leg of the inverter may have corresponding voltage and current sensors.

[0033] Under drive/discharge operating conditions, the power electronics controller **201** controls operation of the electric machine **114**. For example, in response to torque and/or speed setpoints, the power electronics controller **201** may operate the switching devices **210**, **212**, **214**, **216**, **218**, **220** to control the torque and speed of the electric machine **114** to achieve the setpoints. The torque and/or speed setpoints may be processed to generate a desired switching pattern for the switching devices **210**, **212**, **214**, **216**, **218**, **220**. The control terminals of the switching devices **210**, **212**, **214**, **216**, **218**, **220** may be driven with PWM signals to control the torque and speed of the electric machine **114**. The power electronics controller **201** may implement various well-known control strategies to control the electric machine **114** using the switching devices such as vector control and/or six-step control. During discharge operating conditions, the switching devices **210**, **212**, **214**, **216**, **218**, **220** are actively controlled to achieve a desired current through each phase of the electric machine **114**.

[0034] Under regenerative/charge operating conditions (e.g., regenerative mode), the power electronics controller **201** may control the power electronics module **126** to accommodate power generated by the electric machine **114**. For example, the power electronics controller **201** may operate the switching devices **210**, **212**, **214**, **216**, **218**, **220** to convert AC power generated by the electric machine **114** to DC current to charge the traction battery **124** via the HV busbar **152**. The power electronics controller **201** may implement various well-known control strategies to perform the regenerative operation.

[0035] The circuit **200** may further include one or more bus capacitors **260** that are coupled across the positive HV busbar **152***a* and negative HV busbar **152***b* via a DC connection **266**. As illustrated in FIG. **2**, a positive terminal of the DC connection **266** connects the positive HV busbar **152***a* to a first terminal of the bus capacitor **260**, and a negative terminal of the DC connection **266** connects the negative HV busbar **152***b* to a second terminal of the bus capacitor **260**. Here, since the DC connection **266** is directly connected to the HV busbars **152**, voltage on the DC connection **266** may be substantially the same as voltage on the HV busbars **152**. The bus capacitors **260** may smooth the voltage of the DC bus **266** as well as the voltage of the HV busbars **152**.

[0036] As discussed above, the power electronics module **126** may generate EMI during the drive and charge operating conditions. The EMI may affect the operations of the power electronics module **126** as well as other components of the vehicle **112** such as the traction battery **124** and the electric machine **114**. The EMI noises may go out of the power electronics module **126** and affect other external systems. An EMI filter **270** may be provided to reduce and/or suppress the EMI such that the interference with components of the vehicle **112** or other external systems is minimized.

[0037] As illustrated in FIG. **2**, the EMI filter **270** may include a first RC pair having a first capacitor **272***a* and a first resistor **274***a*. The first capacitor **272***a* includes a first terminal connected to the positive HV busbar **152***a* and a second terminal connected to ground **276** via the first resistor **274***a*. The first resistor **274***a* includes a first terminal connected to the second terminal of the first capacitor **272***a* and a second terminal connected to the ground **276**. In one example the ground **276** may be implemented as one or more ground busbars **276** that are separated from the HV busbars **152**.

[0038] The EMI filter **270** may include a second RC pair having a second capacitor **272***b* and a second resistor **274***b*. The second capacitor **272***b* includes a first terminal connected to the negative HV busbar **152***b* and a second terminal connected to ground **276** via the second resistor **274***b*. The second resistor **274***b* includes a first terminal connected to the second terminal of the second capacitor **272***b* and a second terminal connected to the ground **276**.

[0039] The EMI filter **270** may include a third RC pair having a third capacitor **272***c* and a third resistor **274***c*. The third capacitor **272***c* includes a first terminal connected to the positive HV busbar **152***a* and a second terminal connected to ground **276** via the third resistor **274***c*. The third resistor **274***c* includes a first terminal connected to the second terminal of the third capacitor **272***c* and a second terminal connected to the ground **276**.

[0040] The EMI filter **270** may include a fourth RC pair having a fourth capacitor **272***d* and a fourth resistor **274***d*. The fourth capacitor **272***d* includes a first terminal connected to the negative HV busbar **152***b* and a second terminal connected to ground **276** via the fourth resistor **274***d*. The fourth resistor **274***d* includes a first terminal connected to the second terminal of the fourth capacitor **272***d* and a second terminal connected to the ground **276**. The first capacitor **272***a*, second capacitor **272***b*, third capacitor **272***c*, and fourth capacitor **272***d* may be referred to as Y-capacitors due to their ground connection characteristics.

[0041] The EMI filter **270** may further include a first inductor **276***a* connected between the first and third RC pair on the positive HV busbar **152***a*, and a second inductor **276***b* connected between the second and fourth RC pair on the negative HV busbar **152***b*. The first inductor **276***a* and second inductor **276***b* may form a common mode choke having the characteristics of blocking high frequency EMI noise while allowing low frequency or DC power to pass through without impedance.

[0042] The EMI filter **270** circuit may be implemented in various manners. Conventionally, the EMI filter **270** circuit is implemented via one or more printed-circuit board (PCB). The present disclosure proposes an EMI filter device and manufacture that does not require a PCB board. More specifically, the present disclosure proposes an EMI filter device that is directly placed inside an inverter system controller (ISC) housing without utilizing a PCB component.

[0043] Referring to FIG. **3** a design diagram **300** of the EMI filter device **270** of one embodiment of the present disclosure. With continuing reference to FIGS. **1** and **2**, the EMI filter device **270** may be placed inside an ISC housing **302** in the present implementation shown in FIG. **3**. The ISC housing **302** may be made of various conductive or non-conductive materials. For instance, the ISC housing **302** may be made of aluminum alloy, magnesium alloy, or the like. Alternatively, the ISC housing **302** may be made of composite material such as fiber glass or plastic. In the present example, the positive HV busbar **152***a* and negative HV busbar **152***b* may be placed in parallel in a longitudinal direction (e.g., x axis) at a central portion of the ISC housing **302**. The positive and negative HV busbars **152** may divide the ISC housing **302** into an upper bank corresponding to the positive HV busbars **152***a* and a lower bank corresponding to the negative HV busbar **152***b*. The HV busbars **152** may be attached to a bottom surface of the ISC housing **302** via one or more support pillars **304** made of insulating material such that the electric power carried by the HV busbars **152** is not transferred to the ISC housing **302**. Alternatively, if the ISC housing **302** is made of non-conductive materials (e.g., fiber glass), the support pillars may be unnecessary and the HV busbars **152** may be directly coupled to an inner surface of the ISC housing **302** without requiring extra insulating elements.

[0044] The ISC housing **302** may be further configured to accommodate one or more Y-capacitors **272** that are attached to the ISC housing **302** in parallel to the HV busbars **152** in the longitudinal direction. The Y-capacitors **272** may be required to insulate from the ISC housing **302**. Thus, an insulating layer may be placed in between the Y-capacitors **272** and the ISC housing **302** if the ISC housing is made of conductive materials. Alternatively, if the ISC housing is made of non-conducting materials, the insulating layer may not be required. The Y-capacitors **272** may be provided with one or more first terminals **306** configured to connect with one of the positive or negative HV busbar **152** as discussed with reference to FIG. **2**. The Y-capacitors **272** may be further provided with one or more second terminals **308** configured to connect with the ground busbar **276** via one or more resistors **274**. In case that the ISC housing **302** is grounded, the ground busbar **276** may be directly connected to the ISC housing **302**. Alternatively, the ground busbar **276** may be insulated from ISC housing **302** and separately grounded.

[0045] The EMI filter device **270** may be further provided with an insulating cover **310** attached to a top surface of the Y-capacitors **272** and configured to provide insulation to the Y-capacitors **272**. The insulating cover **310** may be made of various insulating materials such as fiber glass, plastic, or the like. The insulating cover **310** may be provided with one or more cutout portions **312** contoured accordingly and configured to accommodate the first and second terminals **306**, **308** of the Y-capacitors. As illustrated in FIG. **3**, the terminals **306**, **308** of the Y-capacitors protrude through the top surface of the Y-capacitors for easier connection in the present disclosure. Alternatively, the terminals **306**, **308** may be flush against the top surface of the Y-capacitors in which case the cutouts **312** may be unnecessary for the insulating cover **310**.

[0046] The insulating cover **310** may be configured to of a size that is larger than the top surface area of the Y-capacitors such that the insulating cover **310** may overhang in the transverse direction (e.g., y-axis) toward the HV busbars **152** and/or the ground busbars **276**. The top surface of the insulating cover **310** may be flush against the HV busbars **152** and/or the ground busbars **276** to form a substantially flat top surface together.

[0047] Referring to FIGS. **4**A and **4**B, design diagram **400** and **410** of the EMI filter device with an enlarged portion of the insulating cover are illustrated. The insulating cover **310** may be provided with a notch **402** on an edge near the second terminal **308** of the Y-capacitor **272** and configured to engage a protrusion **412** of the ground busbar **276**. The notch-protrusion engagement design may increase the physical strength of the EMI filter device **270**. The insulating cover **310** may be further provided with a channel **404** connecting the second terminal **308** of the Y-capacitor **272** and configured to accommodate the resistor **274**. The channel **404** does not extend through the entire depth of the insulating cover **310**. Instead, the channel **404** is shallower than the depth of the insulating cover **310** such that the body of the resistor **274** does not directly contact the Y-capacitor **272**. In the present example, the channel **404** is approximately half the depth of the insulating cover **310**. Once installed, the resister **274** is located inside the channel **404** with the first terminal connected to the second terminal **308** of the Y-capacitor **272** and a second terminal connected to the ground busbar **276**. The resistor **274** may be referred to as a surface mount resistor (SMD resistor) **274** due to its mounting location. The height of the resistor **274** may be substantially the same as the depth of the channel **404** such that top surface of the resistor **274** is substantially flush with the insulating cover **310**.

[0048] As illustrated in FIG. **4**B, the length of the resistor **274** may be slightly longer than the length of the channel **404** which makes the second terminal protrude from the insulating cover **310** toward the ground busbar **276**. To accommodate the protruding second terminal of the resistor **274**, the ground busbar **276** may be further provided with a recess **414** at the corresponding location on the protrusion **412**. The notch-protrusion-recess design further increases the physical strength of the EMI filter device **270**. The resistor **274** may be permanently attached to both the second terminal **308** of the Y-capacitor **272** and the ground busbar **276** (e.g., via soldering). Alternatively, the resistor **274** may be removably attached to the second terminal **308** of the Y-capacitor **272** and the ground busbar **276** by friction and/or tension imposed by the channel **404**, the second terminal **308**, and/or the recess **414**.

[0049] Referring to FIG. **5**, a design diagram showing an enlarged portion of the EMI filter of another implementation is illustrated. Different from the example illustrated with references to FIGS. **4**A and **4**B, in the present example, the ground busbar **276** does not include a recess and the resistor **274** is of substantially the same length as the channel **404**.

[0050] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure.

[0051] As previously described, the features of various embodiments may be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes may include, but are not limited to strength, durability, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

Claims

1. A vehicle comprising: an electric machine for propelling the vehicle; a battery for supplying electric power to the electric machine; and an electromagnetic interference filter, connected between the electric machine and battery, including a housing, a capacitor disposed within and insulated from the housing, a ground busbar attached to the housing, a resistor having a first terminal connected to a terminal of the capacitor and a second terminal connected to the ground busbar, and an insulating cover at least partially covering a top surface of the capacitor.

2. The vehicle of claim 1, wherein the insulating cover defines a notch, and the ground busbar defines a protrusion configured to engage the notch.

3. The vehicle of claim 2, wherein the ground busbar further defines a recess on the protrusion configured to accommodate a portion of the resistor.

4. The vehicle of claim 1, wherein the terminal of the capacitor protrudes away from the top surface, and the insulating cover defines a cutout configured to accommodate the terminal of the capacitor.

5. The vehicle of claim 1, wherein the insulating cover defines a channel between the terminal of the capacitor and ground busbar and configured to accommodate the resistor.

6. The vehicle of claim 5, wherein a depth of the channel is less than a depth of the insulating

cover.
7. The vehicle of claim 1, wherein the housing is made of conductive material, and the electromagnetic interference filter further includes an insulation layer disposed between the capacitor and housing.
8. The vehicle of claim 1, wherein the resistor is a surface mount resistor.
9. An automotive system comprising: automotive power electronics including an electromagnetic interference filter, that lacks a printed circuit board, disposed within a housing of an inverter system controller.
10. The automotive system of claim 9, wherein the electromagnetic interference filter includes a Y-capacitor and a ground busbar each mounted to the housing, a non-conductive cap plate covering the Y-capacitor, and a surface mount resistor embedded in the non-conductive cap plate and in contact with a terminal of the capacitor and the ground busbar.
11. The automotive system of claim 10 wherein the non-conductive cap plate is plastic.
12. The automotive system of claim 10, wherein the electromagnetic interference filter further includes a DC busbar connected with the Y-capacitor.
13. The automotive system of claim 10, wherein the non-conductive cap plate defines a slot configured to accommodate the surface mount resistor.
14. An automotive component comprising: an inverter system controller including a housing and having disposed within the housing a Y-capacitor defining a terminal, a ground busbar, and a surface mount resistor in direct contact with the terminal and ground busbar to form an electromagnetic filter that lacks a printed circuit board.
15. The automotive component of claim 14 further comprising a non-conductive cap plate covering the Y-capacitor, wherein the surface mount resistor is embedded in the non-conductive cap plate.
16. The automotive component of claim 15, wherein the non-conductive cap plate defines a slot configured to accommodate the surface mount resistor.
17. The automotive component of claim 14, wherein the Y-capacitor and ground busbar are mounted to the housing.
18. The automotive component of claim 14 further comprising a DC busbar disposed within the housing and connected with the Y-capacitor.